

C-2 AT-2

Assessment Tool-2

(Class Test)

Name: K. Naga Pradeep

Reg No: 192525242

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Course Name: Data Ware^{house} and Data Mining

Faculty: Dr. K. V. Kanimochi

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Data Mining:

Data Mining is the process of discovering useful patterns, trends and knowledge from large datasets.

Main tasks in data mining tasks:

1) Classification

2) Clustering

Steps in the KDD Process:

- * Data cleaning
- * Data Integration
- * Data Selection
- * Data Transformation
- * Data Mining
- * Pattern Evaluation
- * Knowledge Presentation

Role of Data Preprocessing:

- * Improves data quality
- * Removes missing and inconsistent data
- * Increases mining accuracy and efficiency

(b) Data cleaning:-

Data cleaning is the process of detecting and correcting incomplete or inconsistent data.

Two data quality issues:-

i, Missing values.

ii, Noisy data

Handling Missing Values:-

- Replace with mean/median/mode.
- Predict values using machine learning.

Example:-

Age = ?, replace with average age = 28

Handling Noisy Data:-

- * Binning
- * Regression
- * Clustering

Example:-

Marks: 45, 46, 47, 48 → smooth using binning.

Data Integration:

Data Integration is the process of combining data from different sources into a single, consistent dataset for analysis. It is an important preprocessing step in data mining.

Challenges:-

- * Data Redundancy - Duplicate data stored in different sources.
- * Data Inconsistency - Different formats or conflicting values.

Handling Redundancy:-

- ⇒ Remove duplicate records
- ⇒ Use correlation analysis to identify repeated attributes.

Handling Inconsistency:-

- * Standardize formats such as dates and units.
- * Resolve conflicts using validation rules and data cleaning.

Proper data integration improves consistency, reduces storage and enhances the accuracy of the mining results.

b) Covariance Analysis:

Covariance analysis is a statistical technique used to measure how two variables change together. It helps determine whether two variables have a positive, negative or no relationship.

Types of Covariance:-

* Population Covariance:-

$$CO_v(x, y) = \frac{\sum (x - \mu_x)(y - \mu_y)}{N}$$

* Sample Covariance:-

$$CO_v(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

Interpretation:-

- * Positive covariance: Variables increase together.
- * Negative covariance: One increases while the other decreases.
- * Zero covariance: No relationship.

Covariance is widely used in finance, business analysis and data mining.

The given stock prices are:

Stock A	Stock B
2	5
3	8
5	10
4	11
6	14

From the data, it can be observed that as the value of stock A increases, the value of stock B also generally increases. This indicates that both stocks move in the same direction because they are influenced by the same industry trends.

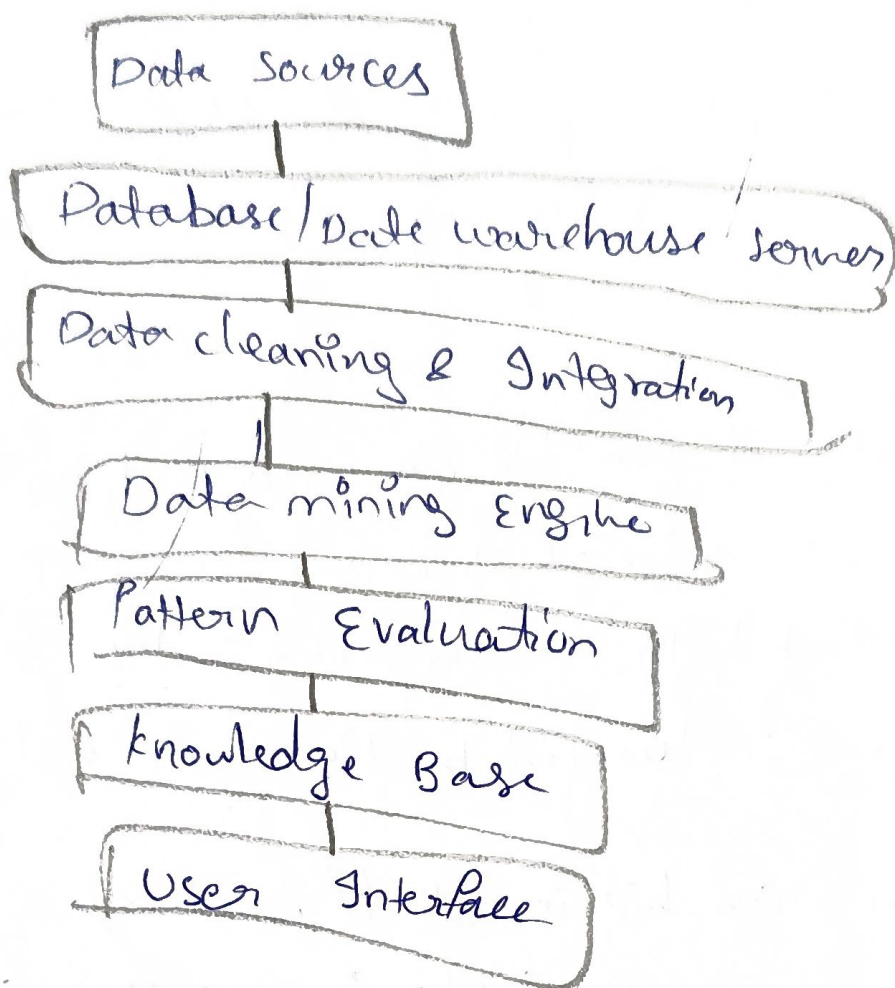
This relationship is called positive covariance, which means that when one variable increases, the other variable also tends to increase. If one stock decreases, the other is also likely to decrease.

Since the prices of both stock change together, they are not independent. Whereas, these stocks show a clear relationship due to common market conditions.

Conclusion:-

- The stock prices rise together
- They have positive covariance.
- They are not independent because both are affected by the same industry trends.

- (b) A typical data mining system consists of several components that work together to discover useful knowledge from data.



Components:-

- * Data Sources :- Databases, data warehouses or files containing raw data.
- * Database server :- Retrieves and manages the required data
- * Data cleaning & Integration module :- Removes errors and combines data
- * Data mining Engine :- Applies algorithms such as classification, clustering and rule mining
- * Pattern Evaluation module :- Evaluates and selects meaningful patterns.
- * Knowledge Base :- Stores domain knowledge and user preferences.

Covariance calculation:

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 5, 7, 5, 6\}$$

$$\text{mean}(x) = 6.2$$

$$\text{mean}(y) = 5.8$$

$$\sum (x - \bar{x})(y - \bar{y}) = -2.8$$

$$\begin{aligned}\text{Sample covariance} &= \frac{-2.8}{4} \\ &= -0.7\end{aligned}$$

$$\therefore \text{Covariance} = -0.7$$

Data Pre-Processing steps:-

Major steps:-

1. Data cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization.

Pipeline:-

- * Remove missing values.
- * Detect and remove outliers.
- * Standardize formats.

* Reduce unnecessary attributes.

Reason:-

This sequence improves data quality before mining.

5.
(a) Given data,

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Total = 27 values.

Divide into 3 bins (9 values each):

* Bin 1:

13, 15, 16, 16, 19, 20, 20, 21, 22

* Bin 2:

22, 25, 25, 25, 25, 30, 33, 33, 35

* Bin 3:

35, 35, 35, 36, 40, 45, 46, 52, 70

Bin-1 mean:-

$$\frac{13+15+16+16+19+20+20+21+22}{9} = \frac{162}{9} = 18$$

Bin-2 mean:-

$$\frac{22+25+25+25+25+30+33+33+35}{9} = \frac{253}{9} \approx 28.11$$

Bin-3 mean:-

$$\frac{55+35+35+36+40+45+46+52+70}{9} = \frac{394}{9} \approx 43.78$$

Smoothed Bin 1:

18, 18, 18, 18, 18, 18, 18, 18, 18

Smoothed Bin 2:

28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11

Smoothed Bin 3:

43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78

Result:-

After applying equal-frequency the data becomes smoother by replacing each values in each bin with their respective mean. This reduces noise & helps in better data analysis.



Data Discretization:

Data Discretization converts continuous values into discrete intervals (bins).

Importance:

- Reduces data complexity.
- Speed up mining.
- Improves classification accuracy.

Equal width Binning	Equal frequency Binning
Equal interval size	Equal no. of records
Easy to implement	Handles skewed data better
may create uneven bins	Balanced distribution

Example:-

Equal width: 0-10, 10-20, 20-30

Equal frequency: each bin contains the same no. of values.

Impact:-

Equal frequency binning usually provides better analysis when the data distribution is uneven.